AI / ML Task 2

1. Objective

The goal of this task was to build an enhanced House Price Prediction system using the California Housing Dataset. The focus was on applying proper data preprocessing, training multiple regression models, evaluating their performance objectively, and selecting the best-performing algorithm based on measurable metrics. This workflow mirrors professional ML engineering practices.

ML Pipeline Overview

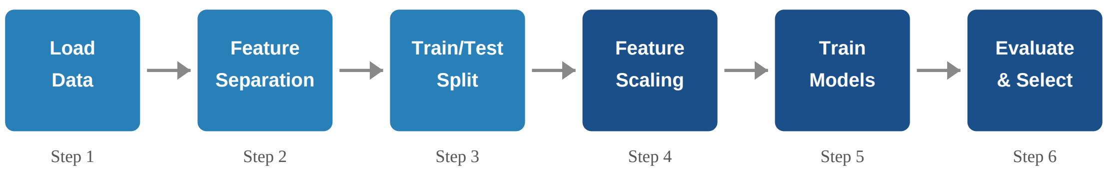

Figure 1 — Six-step ML pipeline followed in this project (Steps 4–6 highlighted).

2. Dataset

The **California Housing Dataset** (sourced via scikit-learn) was used. It contains **20,640 samples** and **8 input features** with no missing values.

Feature	Description
MedInc	Median income in the block group
HouseAge	Median age of houses in the block
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude/Longitude	Geographic location coordinates

Target variable: **Median House Value (Price)** — in units of $100,000.

3. Methodology

Step 1 — Data Loading & Inspection

The dataset was loaded using fetch_california_housing() and structured as a pandas DataFrame. Column names, data types, and missing values were verified; no null entries were found.

Step 2 — Feature & Target Separation

The 8 input features (X) were separated from the target variable Price (y) to prepare the data for supervised learning.

Step 3 — Train-Test Split

The data was split 80/20 (training: 16,512 | test: 4,128 samples) with random_state=42 to ensure reproducibility.

Step 4 — Feature Scaling

StandardScaler was fit on training data only and applied to both sets, preventing data leakage. Scaling is critical here because features span vastly different numeric ranges.

Step 5 — Model Training

Three algorithms were trained: Linear Regression (baseline), Ridge Regression (alpha=1.0, regularization), and Decision Tree Regressor (scale-invariant, captures non-linearity).

Step 6 — Evaluation & Visualization

Models were scored on the test set using RMSE and R^2. Charts were produced to compare performance and inspect prediction quality.

4. Results

The table below summarises model performance on the held-out test set:

Model	RMSE ↓	R^2 Score ↑	Rank
Linear Regression	0.7456	0.5758	3rd
Ridge Regression	0.7456	0.5758	2nd
Decision Tree Regressor	**0.7037**	**0.6221**	**1st ✓**

Metric interpretation: Lower RMSE = better prediction accuracy. Higher R^2 = more variance explained (1.0 = perfect).

Performance Charts

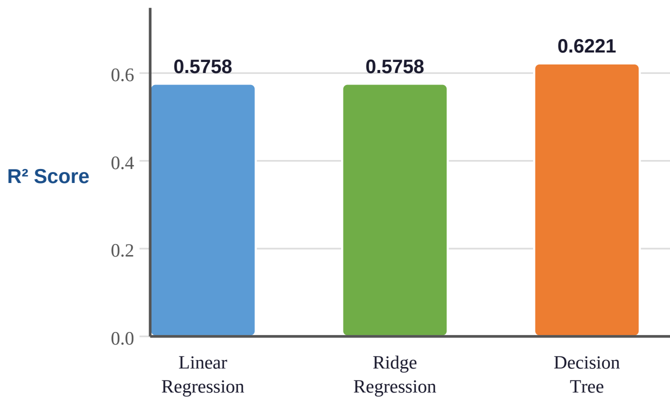

Figure 2 — R^2 Score by model (higher is better).

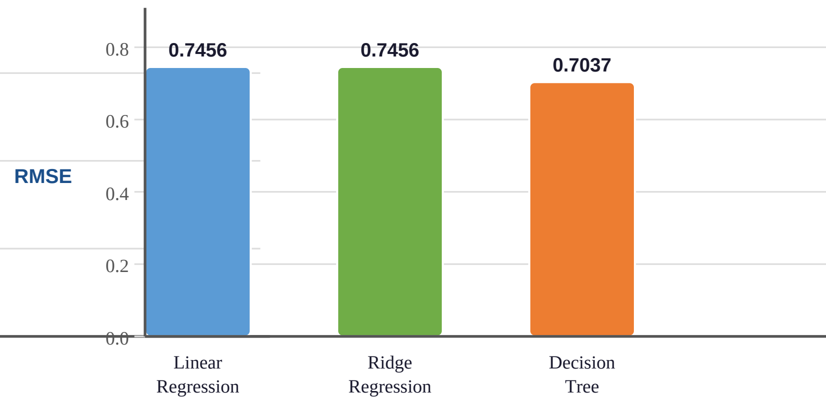

Figure 3 — RMSE by model (lower is better).

Actual vs. Predicted — Linear Regression

Figure 4 — Each point is a test sample; the red diagonal is the perfect-prediction line. Scatter above/below reflects the model's $R^2 \approx 0.576$.

5. Conclusion & Model Selection

The **Decision Tree Regressor** emerged as the best-performing model with the lowest RMSE (0.7037) and the highest R^2 Score (0.6221), explaining approximately **62.2% of the variance** in house prices — a meaningful improvement over both linear models.

Linear Regression and Ridge Regression performed identically ($R^2 \approx 0.576$), confirming that regularization via Ridge offered no benefit on this clean dataset. The Decision Tree's advantage comes from its ability to model **non-linear interactions** between features such as geographic location, income, and occupancy.

Future improvements could include tree pruning (max_depth, min_samples_leaf), ensemble methods such as Random Forest or Gradient Boosting, or cross-validation for more robust evaluation.